
Chapter 3

Circuits and Transmission

Chapter 3 Outline

3.1 - Introduction

3.2 - Circuits

- Configuration, Data Flow, Multiplexing (FDM, TDM, STDM, Inverse Mux, WDM), DSL

3.3 - Communication Media

- Guided and wireless media, media selection

3.4 - Digital Transmission of Digital Data

- Coding, Transmission Modes, Ethernet

3.5 - Analog Transmission of Digital Data (D to A)

- Modulation, Circuit Capacity, Modems

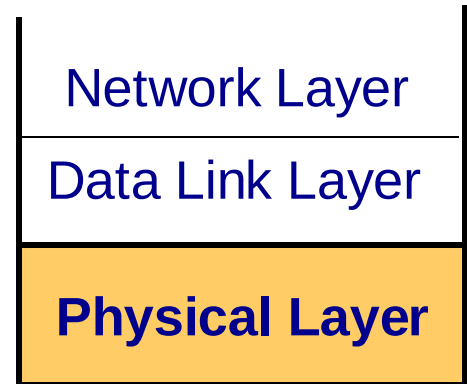
3.6 - Digital Transmission of Analog Data (A to D)

- Translating, Voice Data Transmission, Instant Messenger Transmitting Voice Data, VOIP

3.7 – Implications for Management

3.1 Introduction

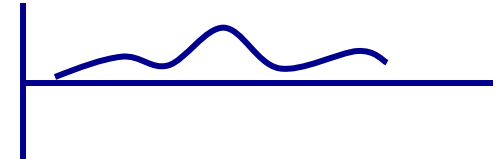
- Includes network hardware and circuits
- Network circuits:
 - physical media (e.g., cables) and
 - special purposes devices (e.g., routers and hubs).
- Types of Circuits
 - Physical circuits connect devices & include actual wires such as twisted pair wires
 - Logical circuits refer to the transmission characteristics of the circuit, such as a T-1 connection refers to 1.544 Mbps
 - Physical and logical circuits may be the same or different. For example, in multiplexing, one physical wire may carry several logical circuits.



Types of Data Transmitted

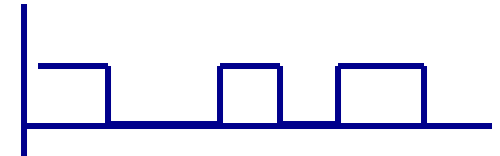
- **Analog data**

- Produced by telephones
- Sound waves, which vary continuously over time, *analogous* to one's voice
- Can take on any value in a wide range of possibilities



- **Digital data**

- Produced by computers, in binary form
- Information is represented as code in a series of ones and zeros
- All digital data is either on or off, 0 or 1



Types of Transmission

- **Analog transmissions**
 - Analog data transmitted in analog form
 - Examples of analog data being sent using analog transmissions are broadcast TV and radio
- **Digital transmissions**
 - Made of discrete square waves with a clear beginning and ending
 - Computer networks send digital data using digital transmissions
- **Data converted between analog and digital formats**
 - Modem (modulator/demodulator): used when digital data is sent as an analog transmission
 - Codec (coder/decoder): used when analog data is sent via digital transmission

Data Type vs. Transmission Type

	<u>Analog Transmission</u>	<u>Digital Transmission</u>
<u>Analog Data</u>	AM and FM Radio, Broadcast TV	Pulse code modulation, MP3, CDs, iPOD, cellphones, VoIP
<u>Digital Data</u>	Dial up modem sending email from your house	Codes such as ASCII or EBCDIC run over Ethernet LANs

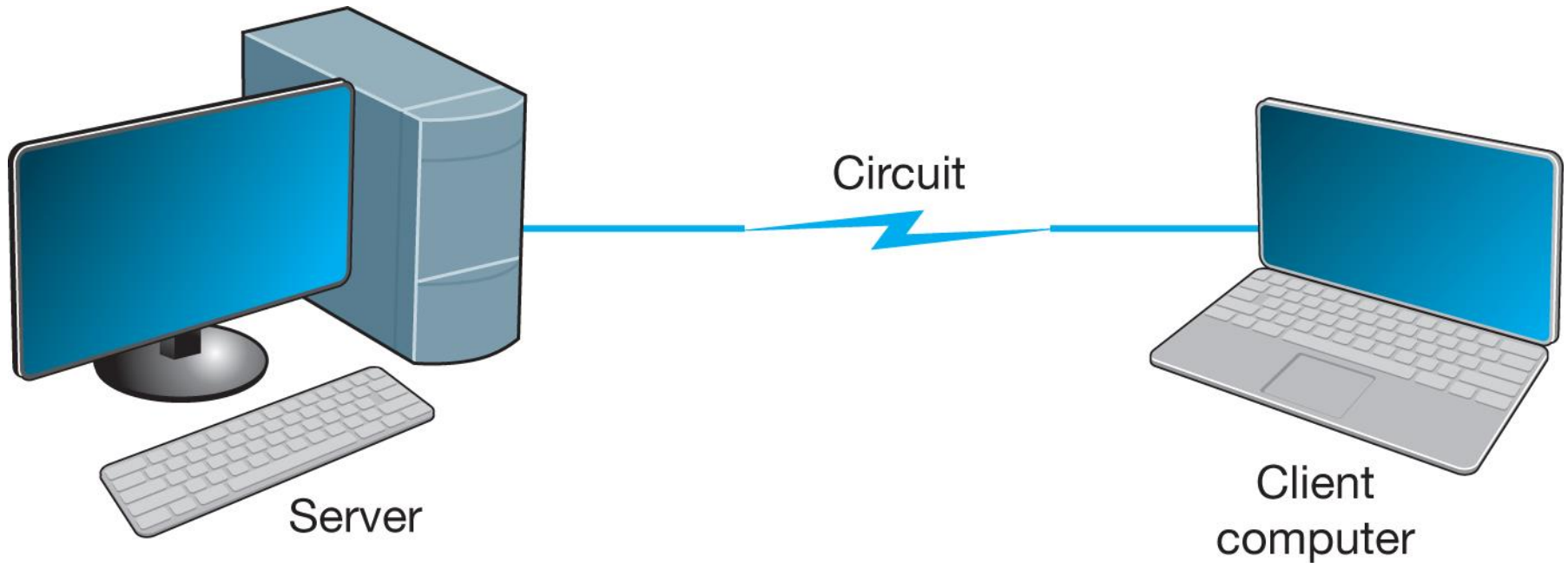
Digital Transmission: Advantages

- **Produces fewer errors**
 - Easier to detect and correct errors, since transmitted data is binary (1s and 0s, only two distinct values)
 - A weak square wave can easily be propagated again in perfect form, allowing more crisp transmission than analog
- **Permits higher maximum transmission rates**
 - e.g., Optical fiber designed for digital transmission
- **More efficient**
 - Possible to send more digital data through a given circuit, circuit can be “packed”
- **More secure**
 - Easier to encrypt digital bit stream
- **Simpler to integrate voice, video and data**
 - Easier mix and match V, V, D on the same circuit, since all signals made up of 0's and 1's

3.2 Circuits

- **Basic physical layout of the circuit**
- **Configuration types:**
 - **Point-to-Point Configuration**
 - **Goes from one point to another**
 - **Sometimes called “dedicated circuits”**
 - **Multipoint Configuration**
 - **Many computer connected on the same circuit**
 - **Sometimes called “shared circuit”**

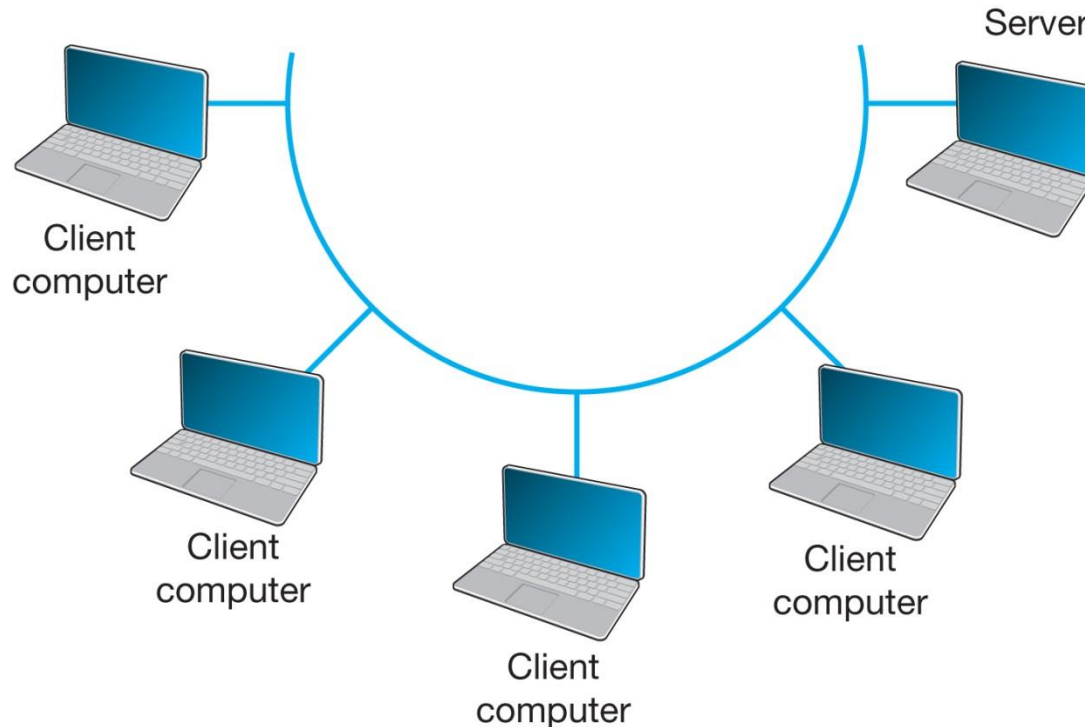
Point-to-Point Configuration



- Used when computers generate enough data to fill the capacity of the circuit
- Each computer has its own circuit to reach the other computer in the network (expensive)

Multipoint Configuration

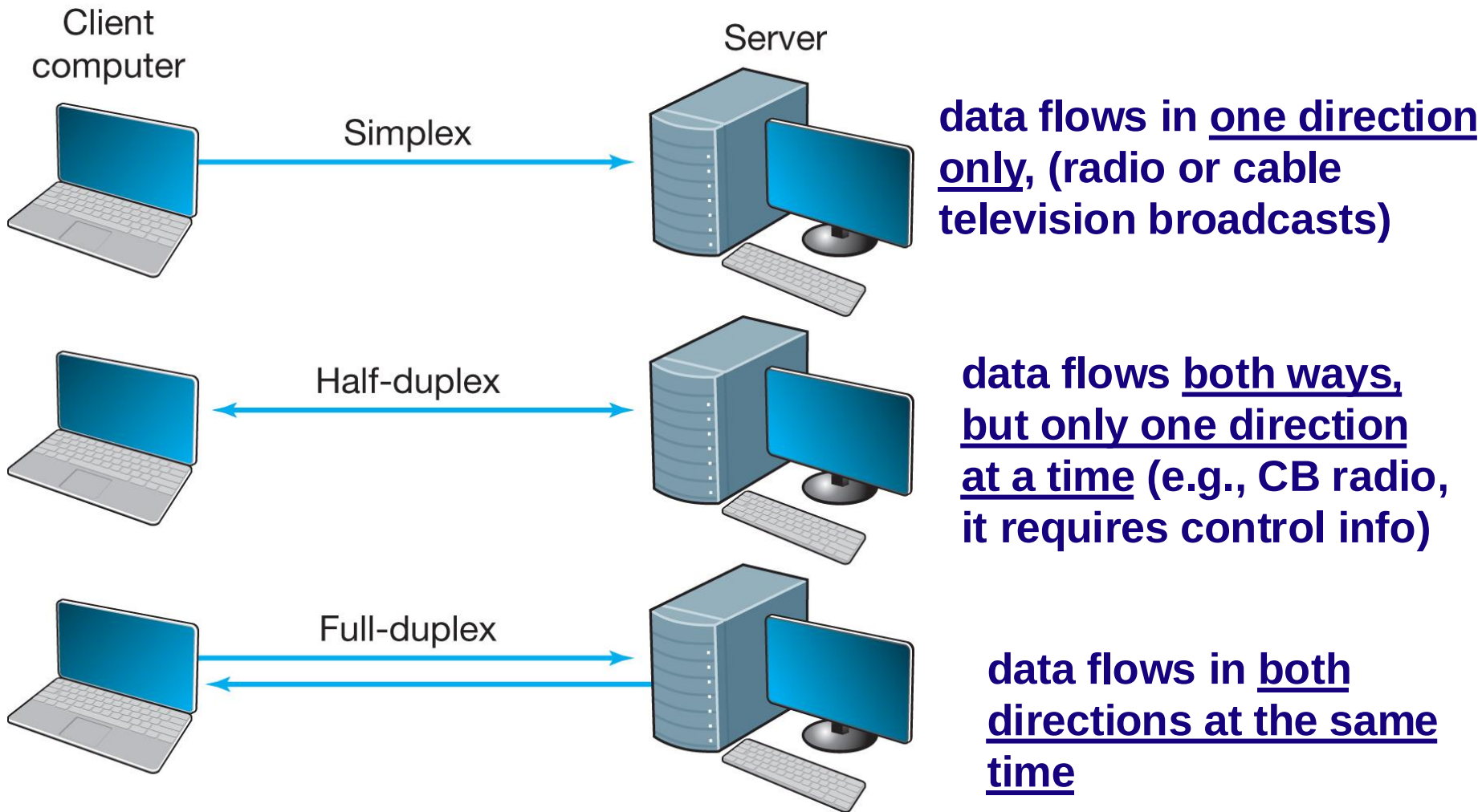
- Used when each computer does not need to continuously use the entire capacity of the circuit



+ Cheaper (not as many wires) and simpler to wire

- Only one computer can use the circuit at a time

Data Flow (Transmission)



Selection of Data Flow Method

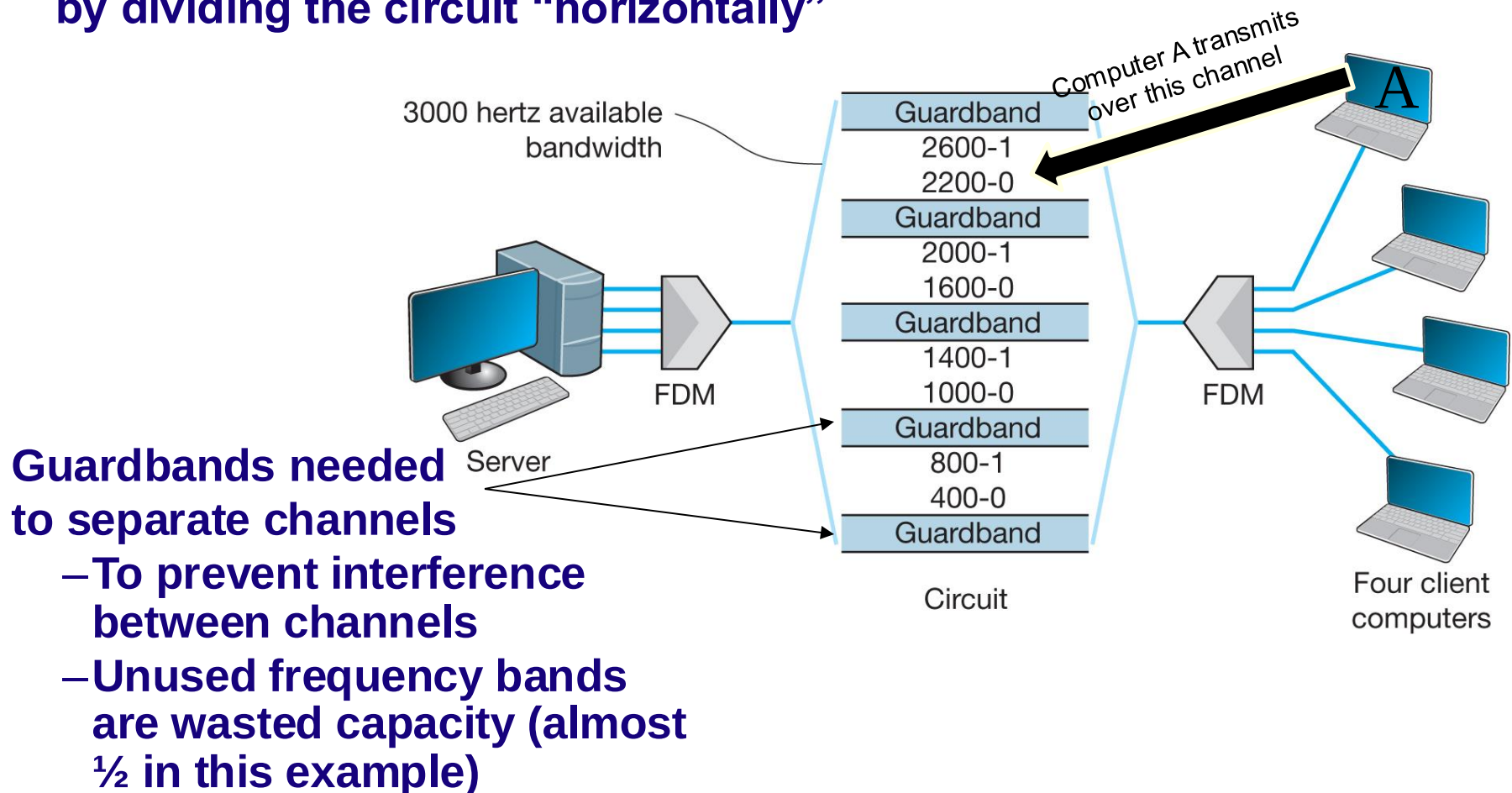
- **Main factor: Application**
 - If data required to flow in one direction only
 - **Simplex Method**
 - e.g., From a remote sensor to a host computer
 - If data required to flow in both directions
 - **Terminal-to-host communication (send and wait type communications)**
 - **Half-Duplex Method**
 - **Client-server; host-to-host communication (peer-to-peer communications)**
 - **Full Duplex Method**
 - **Half-duplex or Full Duplex**
- **Capacity may be a factor too**
 - **Full-duplex uses half of the capacity for each direction**

Multiplexing

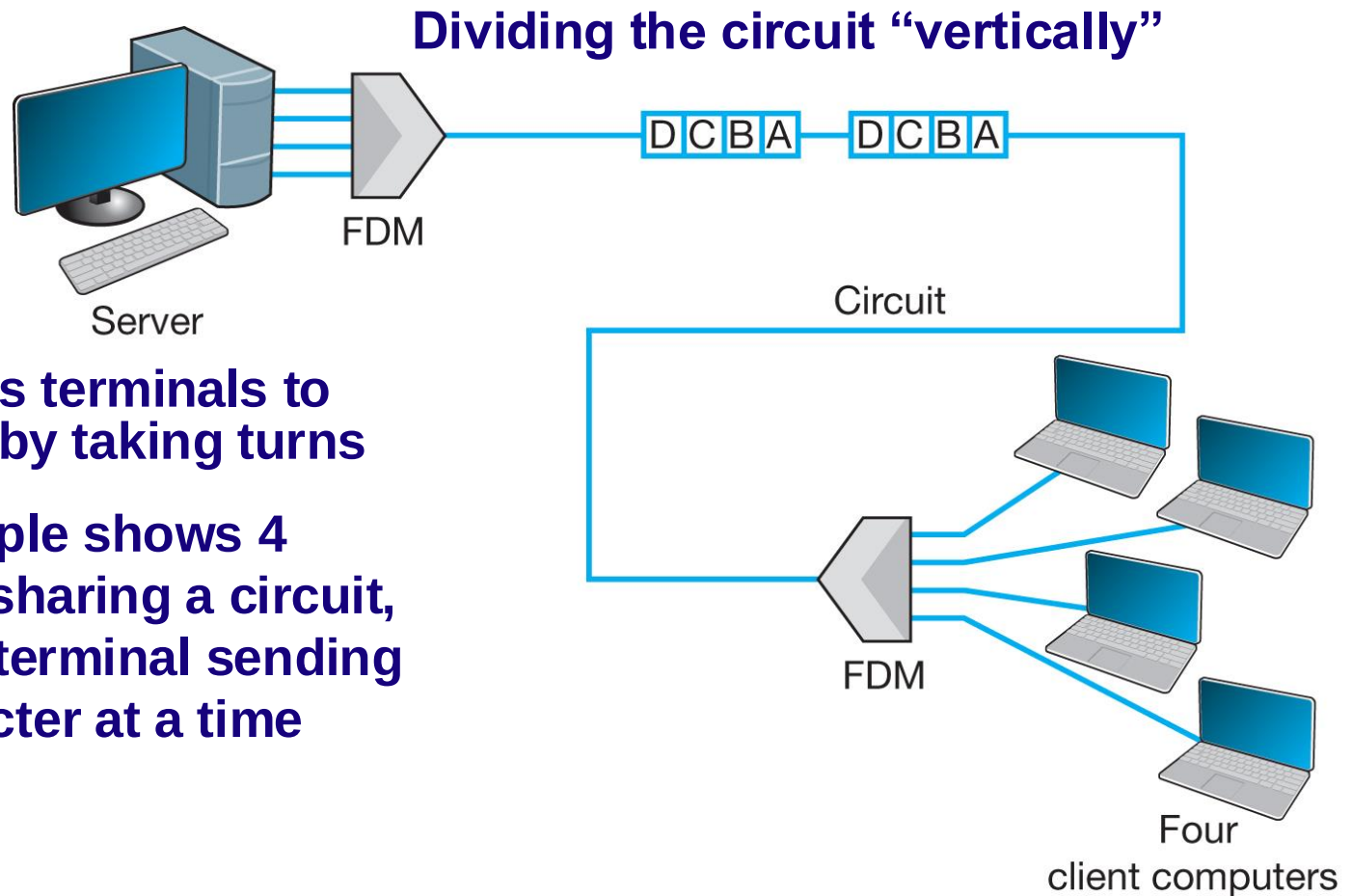
- **Breaking up a higher speed circuit into several slower (logical) circuits**
 - Several devices can use it at the same time
 - Requires two multiplexer: one to combine; one to separate
- **Main advantage: cost**
 - Fewer network circuits needed
- **Categories of multiplexing:**
 - Frequency division multiplexing (FDM)
 - Time division multiplexing (TDM)
 - Statistical time division multiplexing (STDM)
 - Wavelength division multiplexing (WDM)

Frequency Division Multiplexing

Makes a number of smaller channels from a larger frequency band by dividing the circuit “horizontally”



Time Division Multiplexing



Comparison of TDM

- Time on the circuit shared equally
- Each terminal getting a specified timeslot whether needed or not
- More efficient than FDM
- Since TDM doesn't use guardbands, entire capacity can be divided up between terminals

Statistical TDM (STDM)

- **Designed to make use of the idle time slots**
 - In TDM, when terminals are not using the multiplexed circuit, timeslots for those terminals are idle
- **Uses non-dedicated time slots**
 - Time slots used as needed by the different terminals
- **Complexities of STDM**
 - Additional addressing information needed
 - Since source of a data sample is not identified by the time slot it occupies
 - Potential response time delays (when all terminals try to use the multiplexed circuit intensively)
 - Requires memory to store data (in case more data comes in than the outgoing circuit capacity can handle)

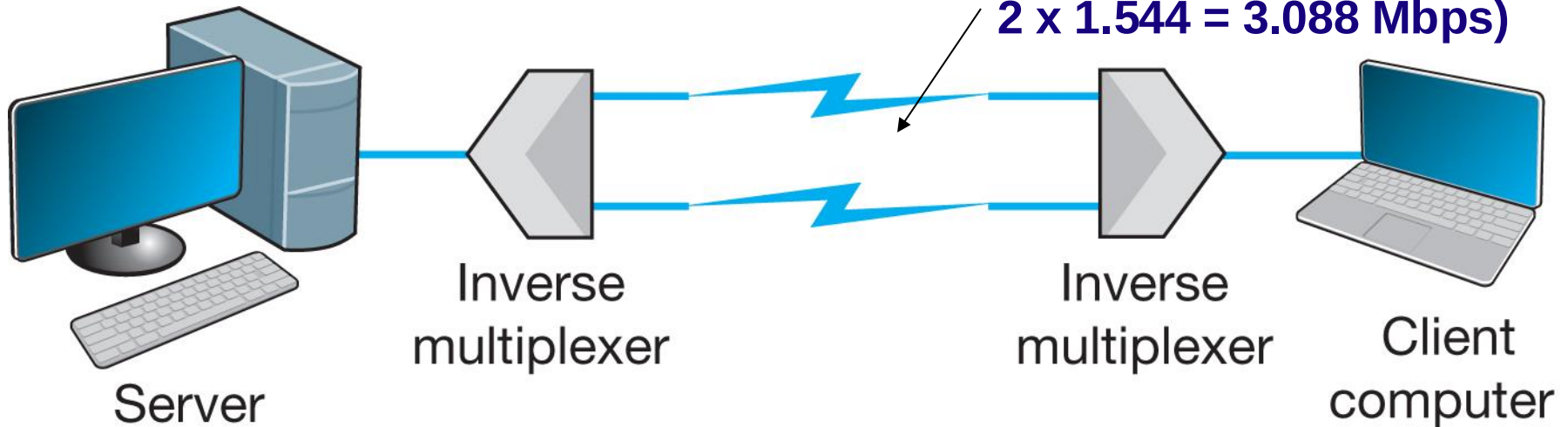
Wavelength Division Multiplexing

- **Transmitting data at many different frequencies**
 - Lasers or LEDs used to transmit on optical fibers
 - Previously single frequency on single fiber (typical transmission rate being around 622 Mbps)
 - Now multi frequencies on single fiber → $n \times 622+$ Mbps
- **Dense WDM (DWDM)**
 - Over a hundred channels per fiber
 - Each transmitting at a rate of 10 Gbps
 - Aggregate data rates in the low terabit range (Tbps)
- **Future versions of DWDM**
 - Both per channel data rates and total number of channels continue to rise
 - Possibility of petabit (Pbps) aggregate rates

Inverse Multiplexing (IMUX)

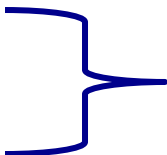
Shares the load by sending data over two or more lines

e.g., two T-1 lines used
(creating a combined
multiplexed capacity of
 $2 \times 1.544 = 3.088$ Mbps)



- **Bandwidth ON Demand Network Interoperability Group (BONDING) standard**
- **Commonly used for videoconferencing applications**
- **Six 64 kbps lines can be combined to create an aggregate line of 384 kbps for transmitting video**

Digital Subscriber Line (DSL)

- Became popular as a way to increase data rates in the local loop.
 - Uses full physical capacity of twisted pair (copper) phone lines (up to 1 MHz) instead of using the 0-4000 KHz voice channel
- 1 MHz capacity split into (FDM):
 - a 4 KHz voice channel
 - an upstream channel
 - a downstream channel

May be divided further (via TDM) to have one or more logical channels
- Requires a pair of DSL modems
 - One at the customer's site; one at the CO site

xDSL

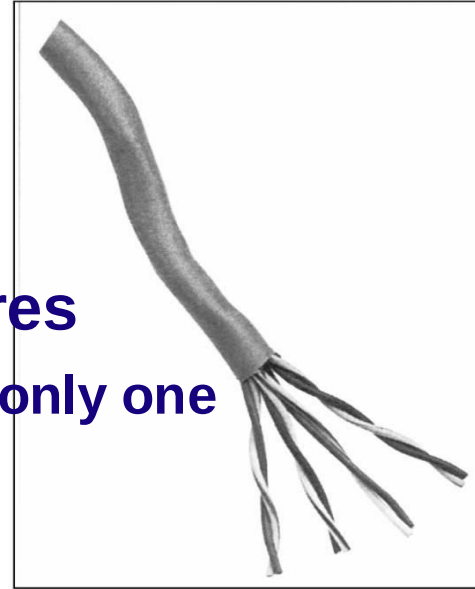
- **Several versions of DSL**
 - Depends on how the bandwidth allocated between the upstream and downstream channels
 - A for Asynchronous, H for High speed, etc
- **G.Lite - a form of ADSL**
 - Provides
 - a 4 Khz voice channel
 - 384 kbps upstream
 - 1.5 Mbps downstream (provided line conditions are optimal).

3.3 Communications Media

- **Physical matter that carries transmission**
- **Guided media:**
 - **Transmission flows along a physical guide (media guides the signal across the network)**
 - **Examples include twisted pair wiring, coaxial cable and fiber optic cable**
- **Wireless media (radiated media)**
 - **No wave guide, the transmission flows through the air or space**
 - **Examples include radio such as microwave and satellite**

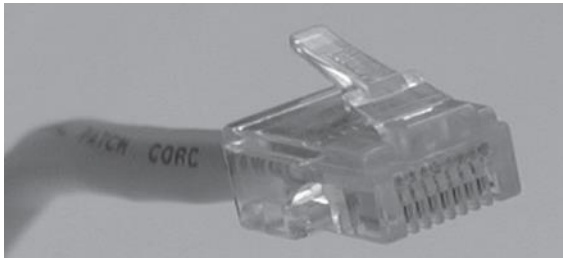
Twisted Pair (TP) Wires

- Commonly used for telephones and LANs
- Reduced electromagnetic interference
 - Via twisting two wires together
(Usually several twists per inch)
- TP cables have a number of pairs of wires
 - Telephone lines: two pairs (4 wires, usually only one pair is used by the telephone)
 - LAN cables: 4 pairs (8 wires)
- Also used in telephone trunk lines (up to several thousand pairs)
- Shielded twisted pair also exists, but is more expensive



Courtesy South Hills Datacomm

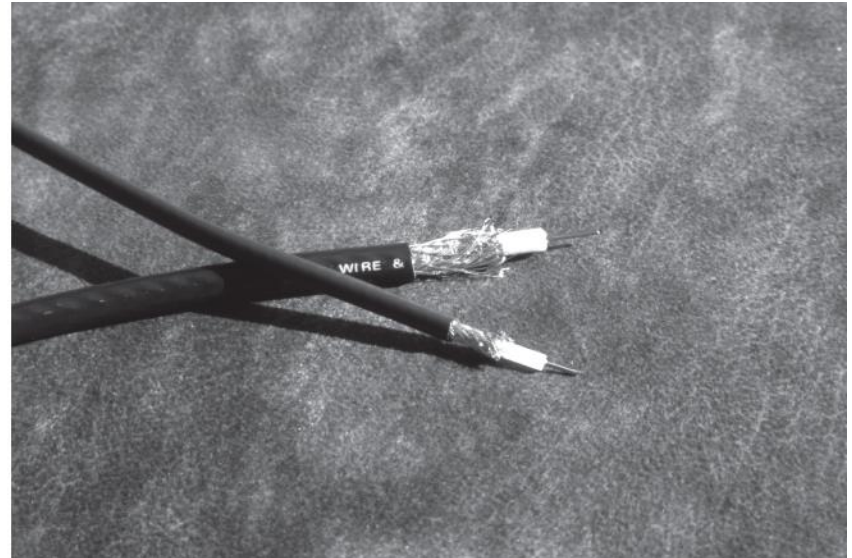
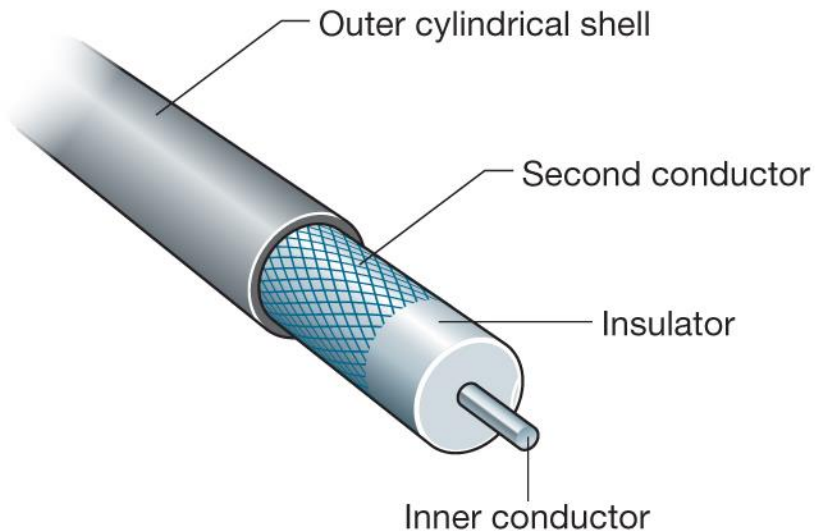
Twisted Pair (TP) Wires



Pin number	Color (EIA/TIA 568B standard)	Name
1	White with orange stripe	Transmit+
2	Orange with white stripe or solid orange	Transmit-
3	White with green stripe	Receive+
4	Blue with white stripe or solid blue	Not used
5	White with blue stripe	Not used
6	Green with white stripe or solid green	Receive-
7	White with brown stripe or solid brown	Not used
8	Brown with white stripe or solid brown	Not used

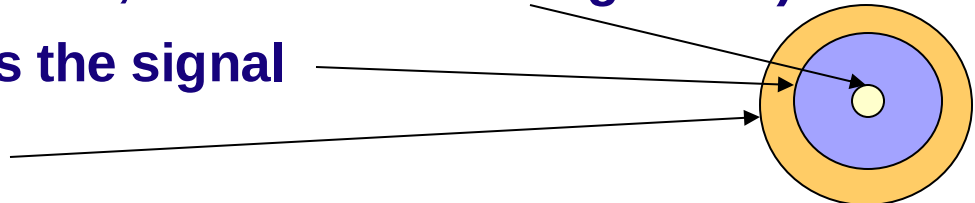
Coaxial Cable

- Less prone to interference than TP due to shielding
- More expensive than TP, thus quickly disappearing
- Used mostly for cable TV



Fiber Optic Cable

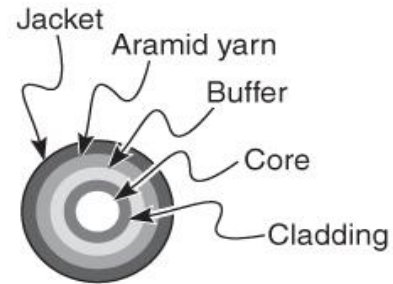
- Light created by an LED (light-emitting diode) or laser is sent down a thin glass or plastic fiber
- Has extremely high capacity, ideal for broadband
- Works well under harsh environments
 - Not fragile, nor brittle; Not heavy nor bulky
 - More resistant to corrosion, fire, water
 - Highly secure, know when is tapped
- Fiber optic cable structure (from center):
 - Core (v. small, 5-50 microns, ~ the size of a single hair)
 - Cladding, which reflects the signal
 - Protective outer jacket



Types of Optical Fiber

- **Multimode (about 50 micron core)**
 - Earliest fiber-optic systems
 - Signal spreads out over short distances (up to ~500m)
 - Inexpensive
- **Graded index multimode**
 - Reduces the spreading problem by changing the refractive properties of the fiber to refocus the signal
 - Can be used over distances of up to about 1000 meters
- **Single mode (about 5 micron core)**
 - Transmits a single direct beam through the cable
 - Signal can be sent over many miles without spreading
 - Expensive (requires lasers; difficult to manufacture)

Optical Fiber



Step index (multimode)



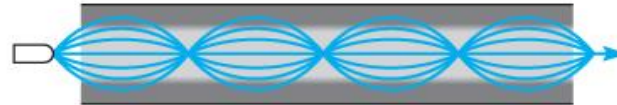
Graded index (multimode)



Single mode



Different parts of signal arrive at different times, signal dispersion



Center light likely to arrive at the same time as the other parts

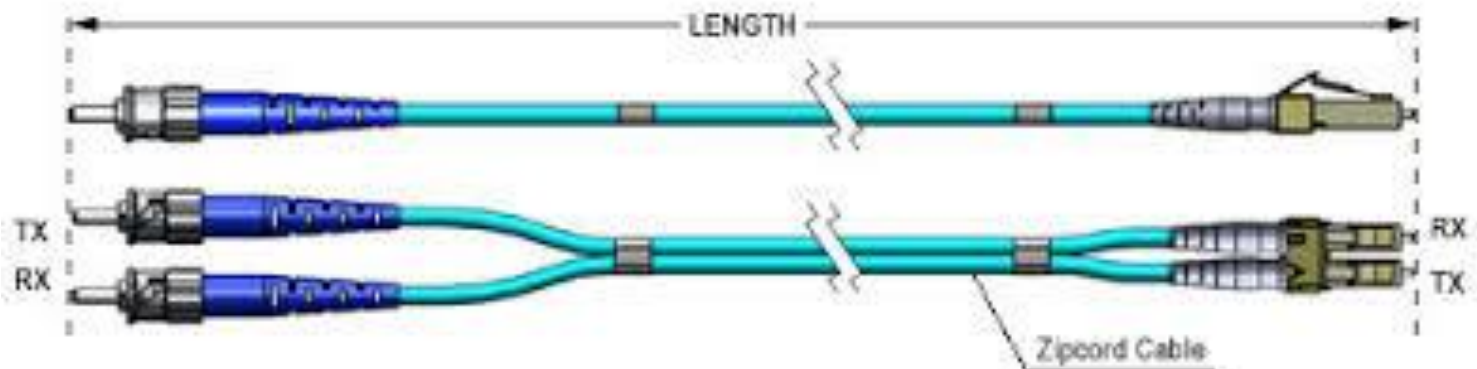


Parallel Fiber Optics

10Gb Duplex
Multimode 50/125
OM3 LSZH Fiber
Patch Cable



Transmitter
Receiver



Radio Waves

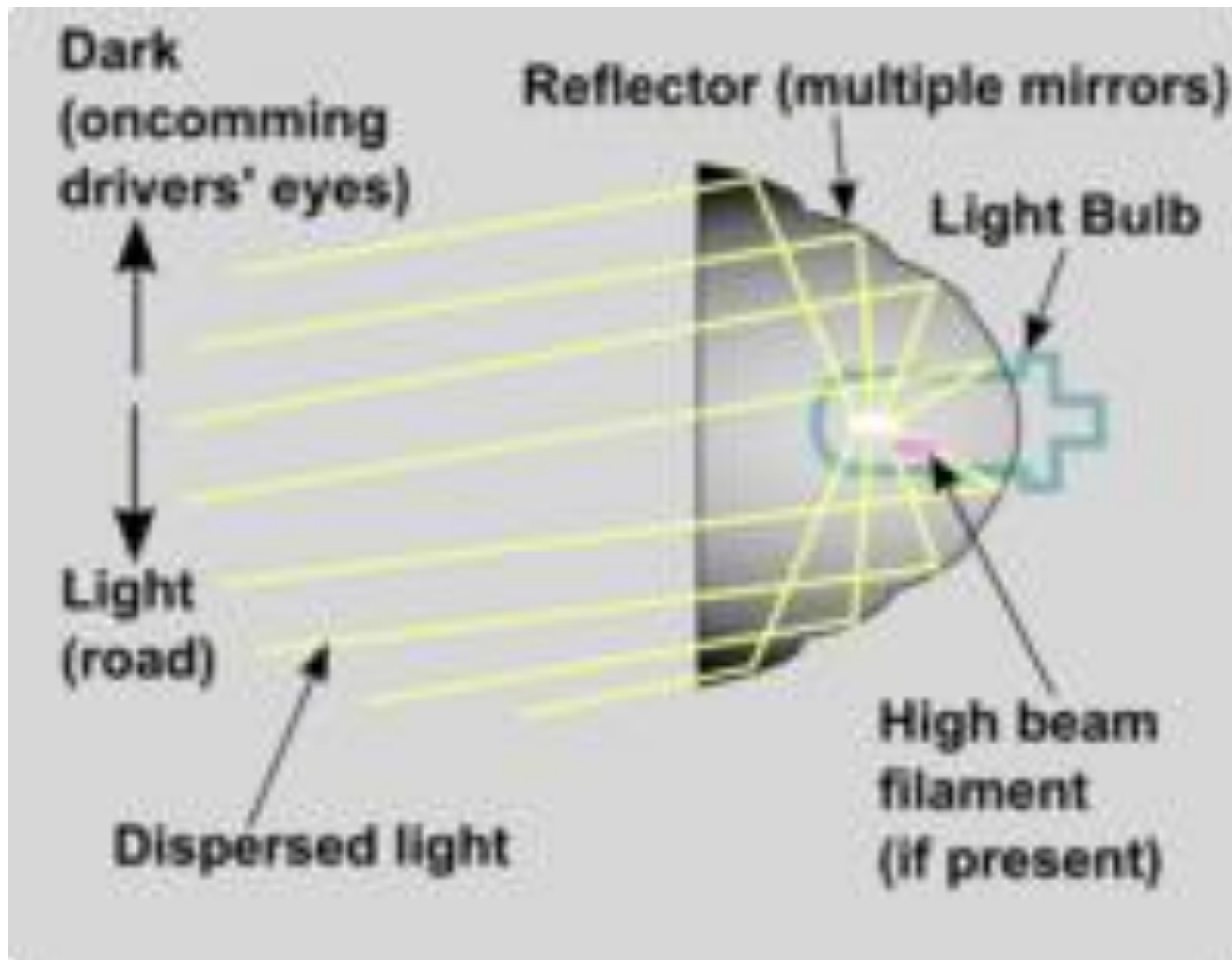
- Wireless transmission of electrical waves through air
- Each device has a radio transceiver with a specific frequency
 - Low power transmitters (few miles range)
 - Often attached to portables (Laptops, PDAs, cell phones)
- Includes
 - AM and FM radios, Cellular phones
 - Wireless LANs (IEEE 802.11) and Bluetooth
 - Microwaves and Satellites, Low Earth Orbiting Satellites

Microwave Radio

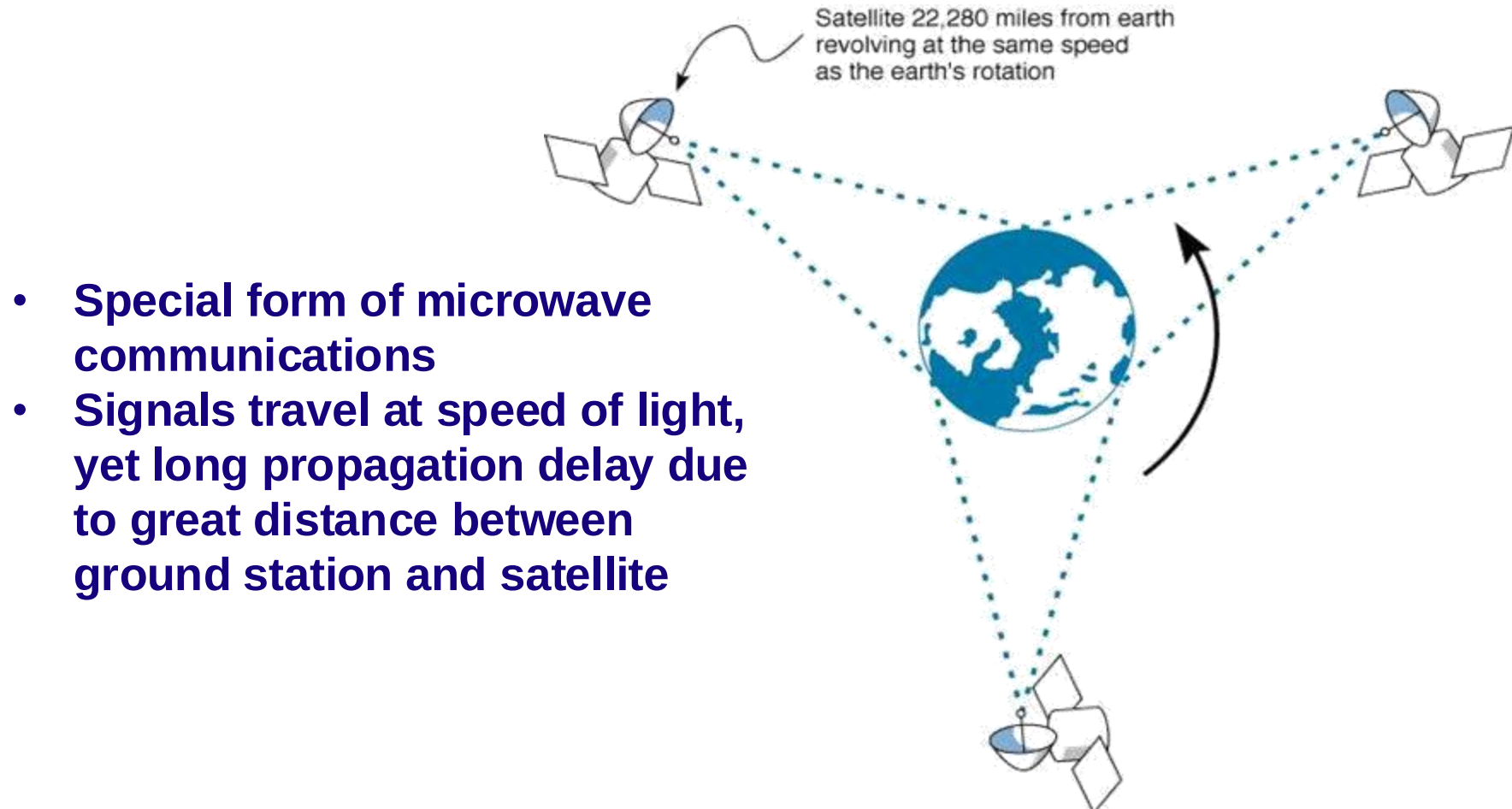
- **High frequency form of radio communications**
 - Extremely short (*micro*) wavelength (1 cm to 1 m)
 - Requires line-of-sight
- **Performs same functions as cables**
 - Often used for long distance, terrestrial transmissions (over 50 miles without repeaters)
 - No wiring and digging required
 - Requires large antennas (about 10 ft) and high towers
- **Possesses similar properties as light**
 - Reflection, refraction, and focusing
 - Can be focused into narrow powerful beams for long distance
 - Some effect from water, rain and snow

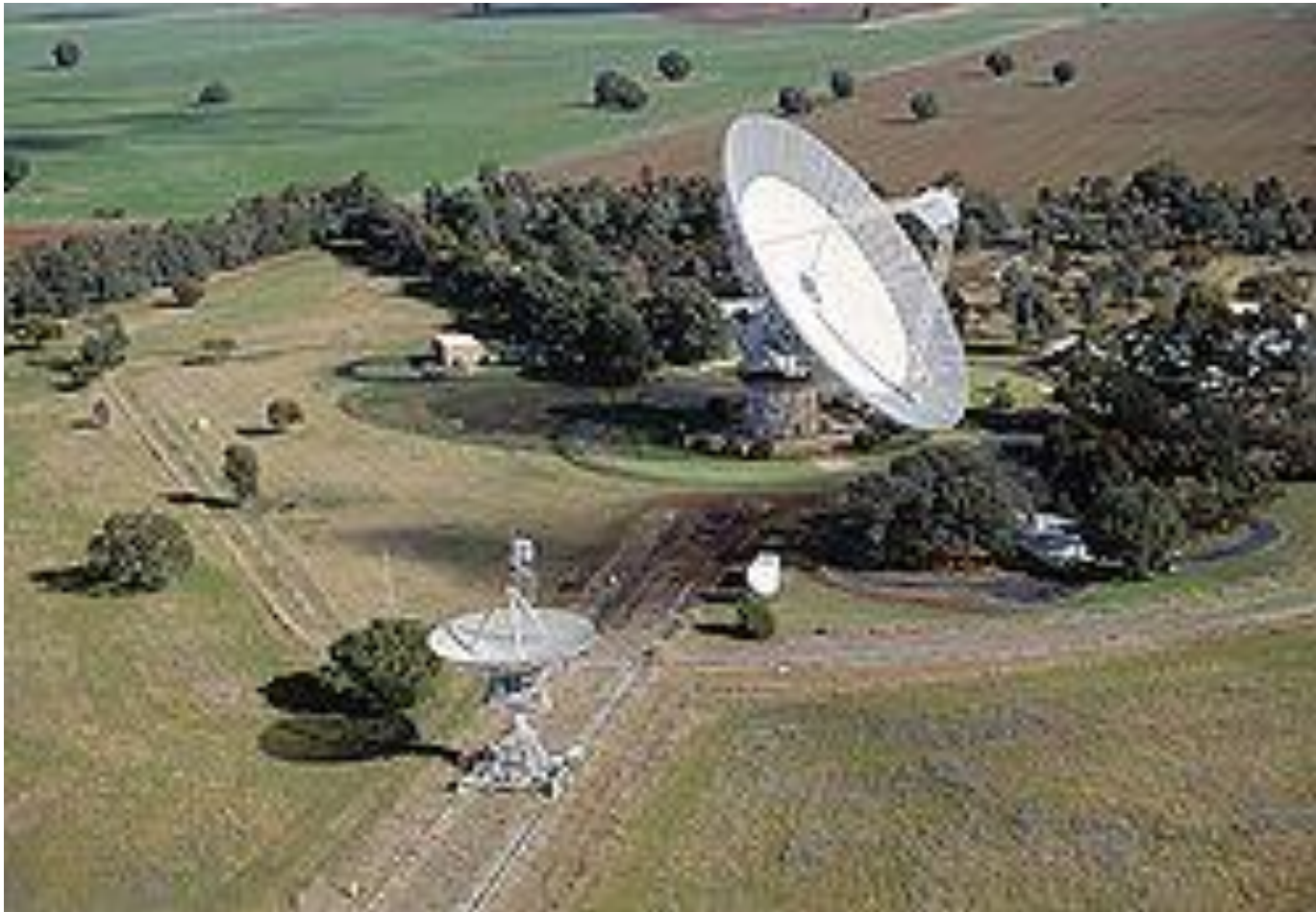


Parabolic Reflector



Satellite Communications







Factors Used in Media Selection

- **Type of network**
 - LAN, WAN, or Backbone
- **Cost**
 - Always changing; depends on the distance
- **Transmission distance**
 - Short: up to 300 m; medium: up to 500 m
- **Security**
 - Wireless media is less secure
- **Error rates**
 - Wireless media has the highest error rate (interference)
- **Transmission speeds**
 - Constantly improving; Fiber has the highest

Media Summary

Guided Media

Media	Network Type	Cost	Transmission Distance	Security	Error Rates	Speed
Twisted Pair	LAN	Low	Short	Good	Low	Low-high
Coaxial Cable	LAN	Moderate	Short	Good	Low	Low-high
Fiber Optics	Any	High	Moderate-long	Very good	Very low	High-very high

Radiated Media

Media	Network Type	Cost	Transmission Distance	Security	Error Rates	Speed
Radio	LAN	Low	Short	Poor	Moderate	Moderate
Microwave	WAN	Moderate	Long	Poor	Low-moderate	Moderate
Satellite	WAN	Moderate	Long	Poor	Low-moderate	Moderate

3.4 Digital Transmission of Digital Data

- Computers produce binary data
- Standards needed to ensure both sender and receiver understands this data
 - *Codes*: digital combinations of bits making up languages that computers use to represent letters, numbers, and symbols in a message
 - *Signals*: electrical or optical patterns that computers use to represent the coded bits (0 or 1) during transmission across media

Coding

- Coding is the representation of a set of characters by a string of bits
 - Letters (A, B, ..), numbers (1, 2,..), special symbols (#, \$, ..)
- ASCII: American Standard Code for Information Interchange
 - Originally used a 7-bit code (128 combinations), but an 8-bit version (256 combinations) is now in use
 - Found on PC computers
- EBCDIC: Extended Binary Coded Decimal Interchange Code
 - An 8-bit code developed by IBM
 - Used mostly in mainframe computer environment

ASCII Chart

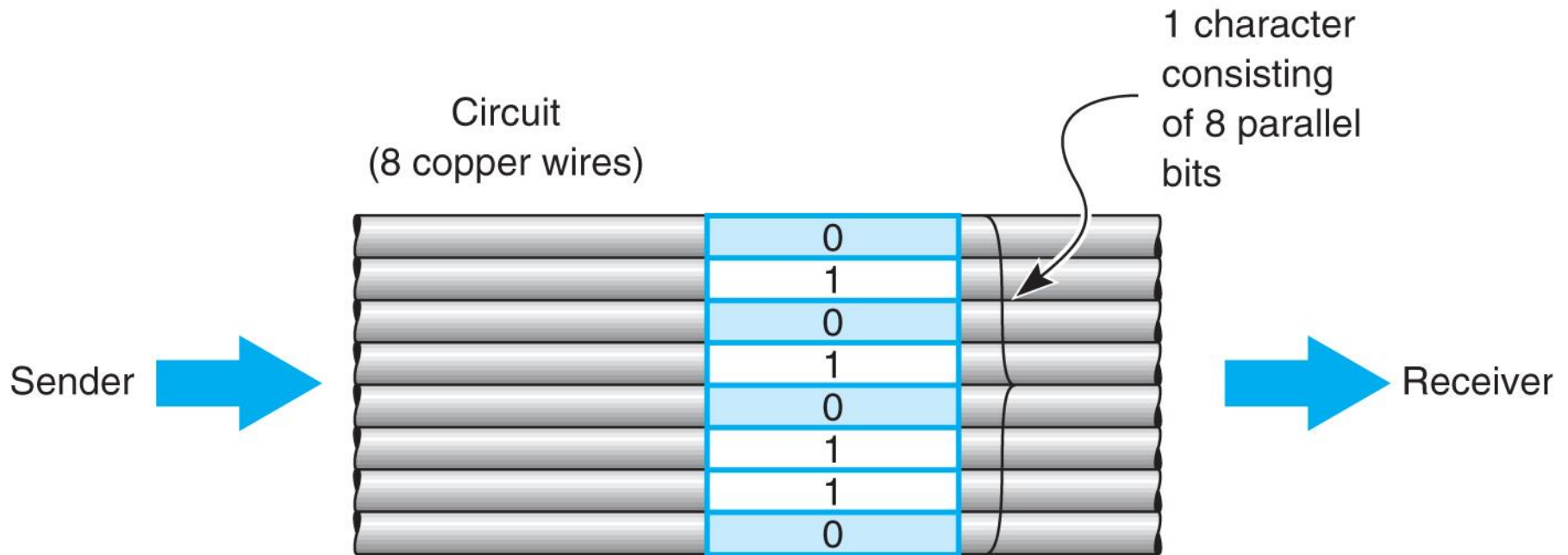
0	0011 0000	O	0100 1111	m	0110 1101
1	0011 0001	P	0101 0000	n	0110 1110
2	0011 0010	Q	0101 0001	o	0110 1111
3	0011 0011	R	0101 0010	p	0111 0000
4	0011 0100	S	0101 0011	q	0111 0001
5	0011 0101	T	0101 0100	r	0111 0010
6	0011 0110	U	0101 0101	s	0111 0011
7	0011 0111	V	0101 0110	t	0111 0100
8	0011 1000	W	0101 0111	u	0111 0101
9	0011 1001	X	0101 1000	v	0111 0110
A	0100 0001	Y	0101 1001	w	0111 0111
B	0100 0010	Z	0101 1010	x	0111 1000
C	0100 0011	a	0110 0001	y	0111 1001
D	0100 0100	b	0110 0010	z	0111 1010
E	0100 0101	c	0110 0011	.	0010 1110
F	0100 0110	d	0110 0100	,	0010 0111
G	0100 0111	e	0110 0101	:	0011 1010
H	0100 1000	f	0110 0110	;	0011 1011
I	0100 1001	g	0110 0111	?	0011 1111
J	0100 1010	h	0110 1000	!	0010 0001
K	0100 1011	I	0110 1001	'	0010 1100

Transmission Modes

- **Bits in a message can be sent on:**
 - a single wire one after another (Serial transmission)
 - multiple wires simultaneously (Parallel transmission)
- **Serial Mode**
 - Sends bit by bit over a single wire
 - Serial mode is slower than parallel mode
- **Parallel mode**
 - Uses several wires, each wire sending one bit at the same time as the others
 - A parallel printer cable sends 8 bits together
 - Computer's processor and motherboard also use parallel busses (8 bits, 16 bits, 32 bits) to move data around

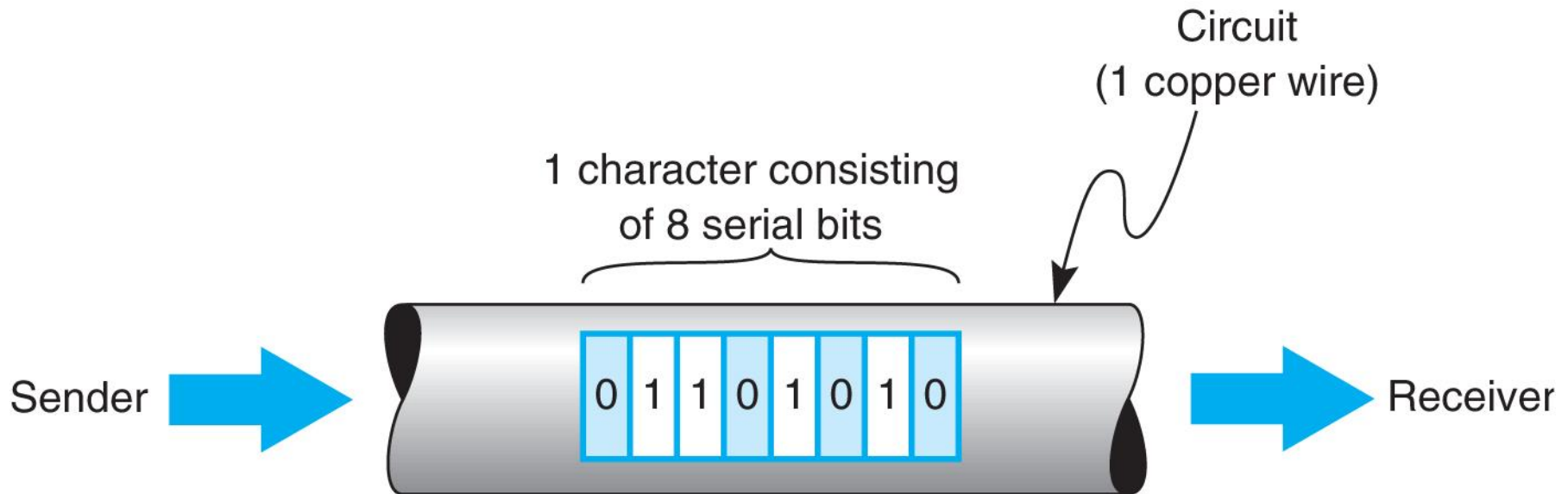
Parallel Transmission Example

Used for short distances (up to 6 meters)
since bits sent in parallel mode tend to
spread out over long distances



Serial Transmission Example

Can be used over longer distances
since bits stay in the order they were
sent



Signaling of Bits

- **Digital Transmission**
 - Signals sent as a series of “square waves” of either positive or negative voltage
 - Voltages vary between +3/-3 and +24/-24 depending on the circuit
- **Signaling (encoding)**
 - Defines how the voltage levels will correspond to the bit values of 0 or 1
 - Examples:
 - Unipolar, Bipolar
 - RTZ, NRZ, Manchester
 - Data rate: describes how often the sender can transmit data
 - 64 Kbps → once every 1/64000 of a second

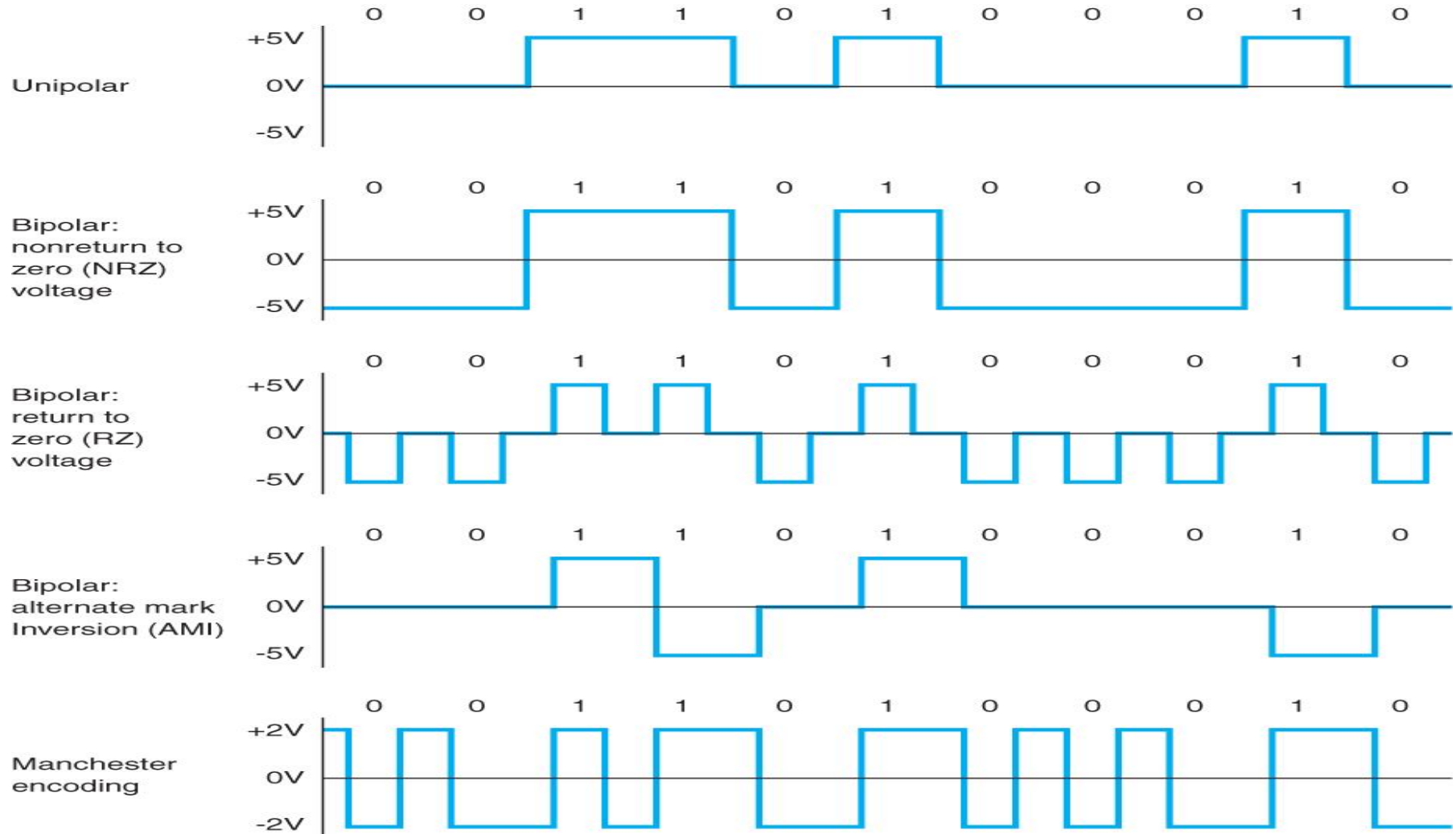
Signaling (Encoding) Techniques

- **Unipolar signaling**
 - Use voltages either vary between 0 and a positive value or between 0 and some negative value
- **Bipolar signaling**
 - Use both positive and negative voltages
 - Experiences fewer errors than unipolar signaling
 - Signals are more distinct (more difficult for interference to change polarity of a current)
 - Return to zero (RZ)
 - Signal returns to 0 voltage level after sending a bit
 - Non return to zero (NRZ)
 - Signals maintains its voltage at the end of a bit
 - Manchester encoding (used by Ethernet)

Manchester Encoding

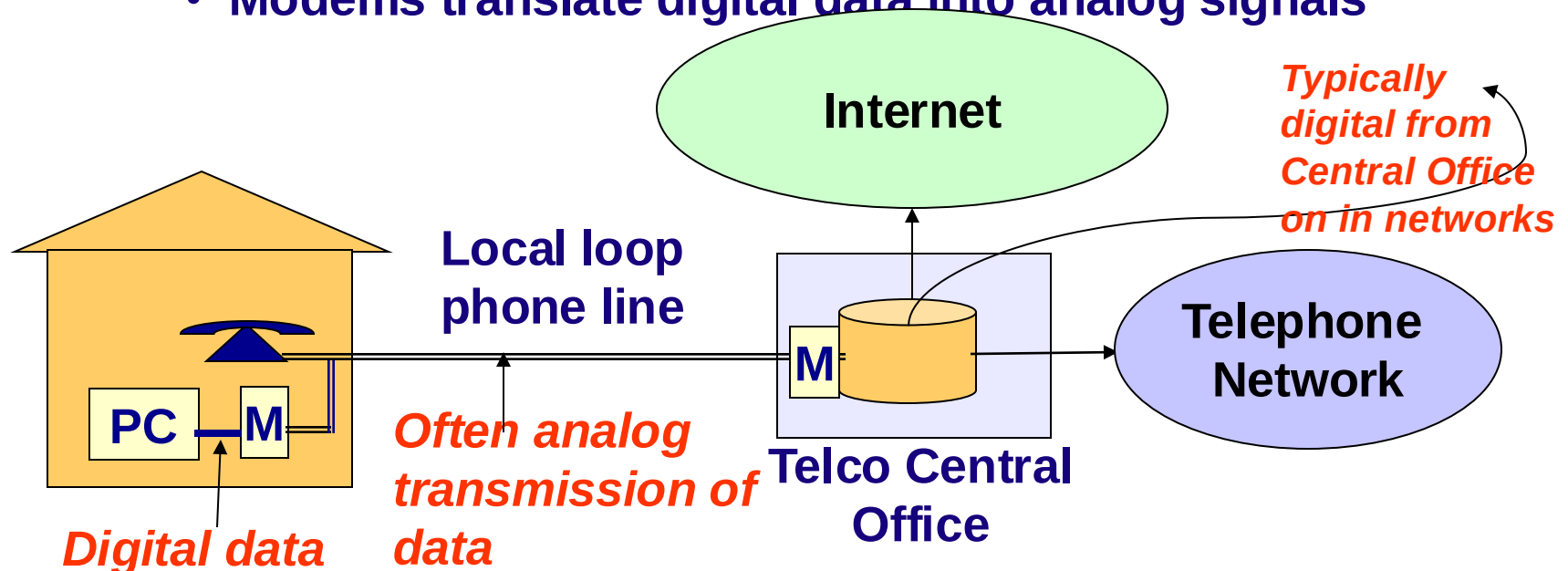
- **Used by Ethernet, most popular LAN technology**
- **Defines a bit value by a mid-bit transition**
 - A high to low voltage transition is a 0 and a low to high mid-bit transition defines a 1
- **Data rates: 10 Mb/s, 100 Mb/s, 1 Gb/s**
 - 10- Mb/s → one signal for every 1/10,000,000 of a second (10 million signals or bits every second)
- **Less susceptible to having errors go undetected**
 - If there is no mid-bit voltage transition, then an error took place

Digital Transmission Types



3.5 Analog Transmission of Digital Data

- A well known example using phone lines to connect PCs to the Internet
 - PCs generate digital data
 - Local loop phone lines use analog transmission technology
 - Modems translate digital data into analog signals

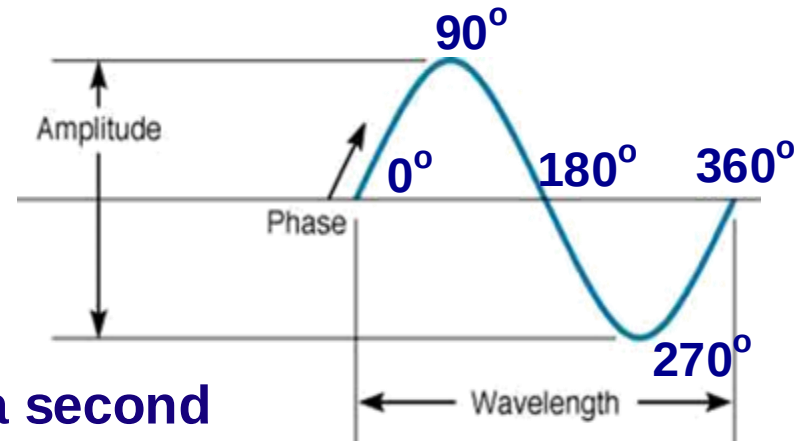


Telephone Network

- Originally designed for human speech (analog communications) only
- POTS (Plain Old Telephone Service)
 - Enables voice communications between two telephones
 - Human voice (sound waves) converted to electrical signals by the sending telephone
 - Signals travel through POTS and converted back to sound waves at far end
- Sending digital data over POTS
 - Use modems to convert digital data to an analog format
 - One modem used by sender to produce analog data
 - Another modem used by receiver to regenerate digital data

Sound Waves and Characteristics

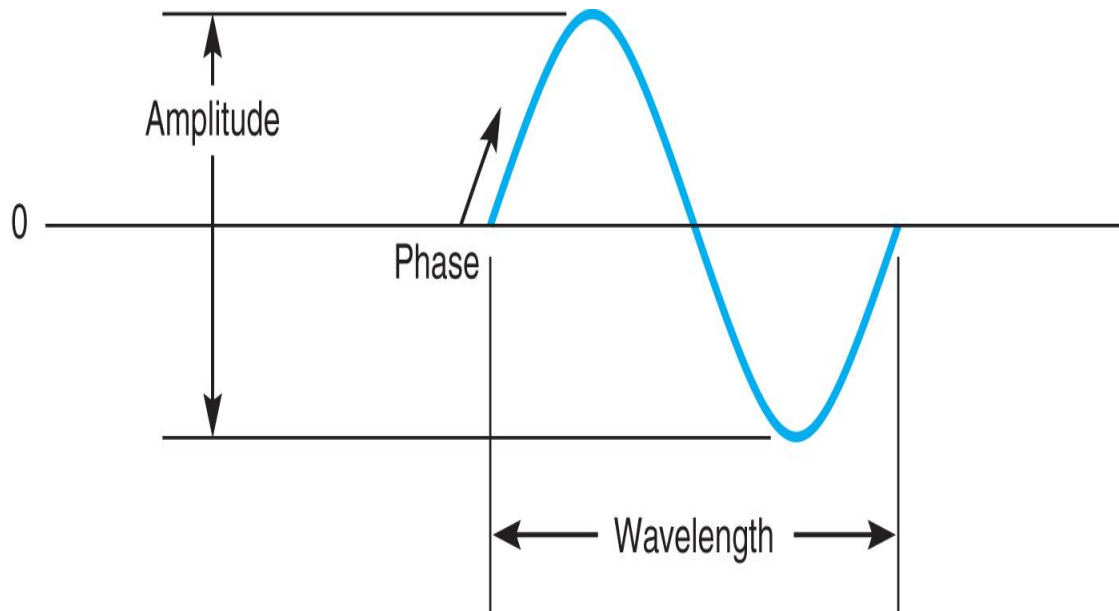
- **Amplitude**
 - Height (loudness) of the wave
 - Measured in decibels (dB)
- **Frequency:**
 - Number of waves that pass in a second
 - Measured in Hertz (cycles/second)
 - Wavelength, the length of the wave from crest to crest, is related to frequency
- **Phase:**
 - Refers to the point in each wave cycle at which the wave begins (measured in degrees)
 - (For example, changing a wave's cycle from crest to trough corresponds to a 180 degree phase shift).



Wavelength vs. Frequency

speed = frequency * wavelength

$$v = f \lambda$$



$$\begin{aligned} v &= 3 \times 10^8 \text{ m/s} \\ &= 300,000 \text{ km/s} \\ &= 186,000 \text{ miles/s} \end{aligned}$$

Example:

if $f = 900 \text{ MHz}$

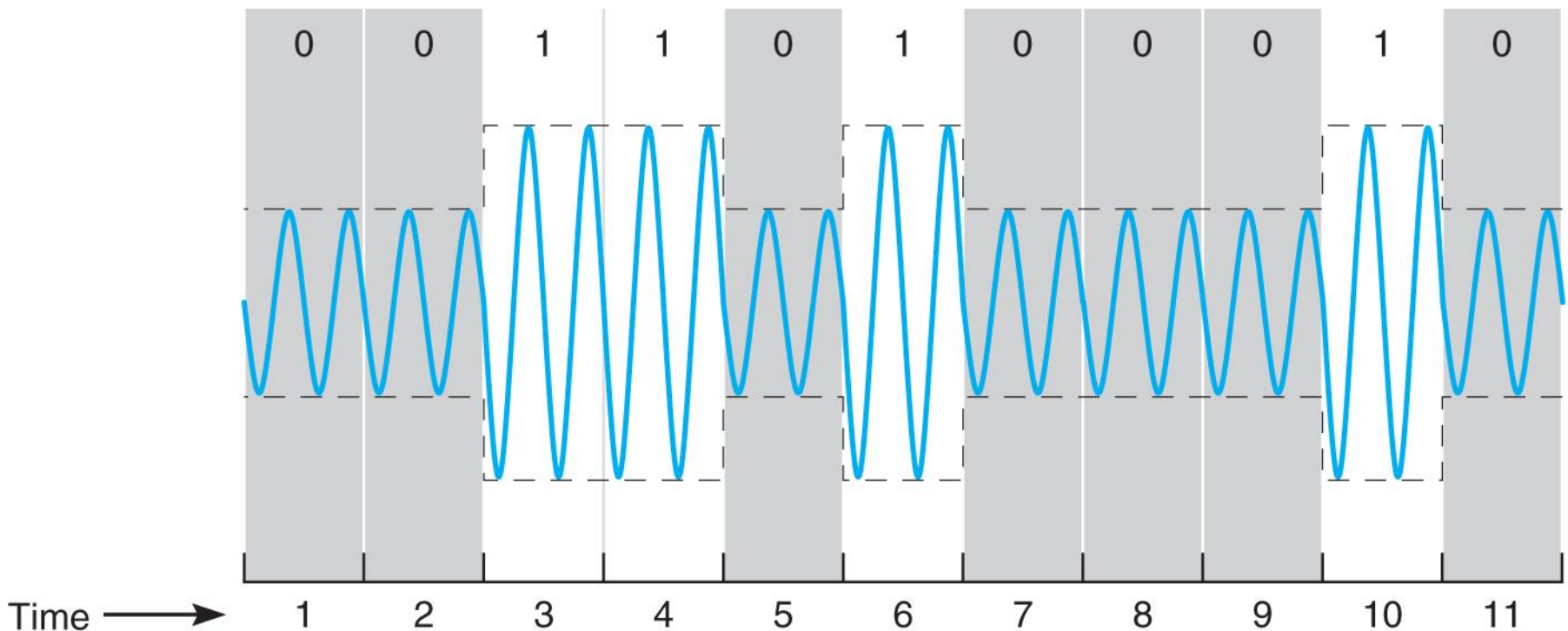
$$\begin{aligned} \lambda &= 3 \times 10^8 / 900 \times 10^3 \\ &= 3/9 = 0.3 \text{ meters} \end{aligned}$$

Modulation

- **Modification of a carrier wave's fundamental characteristics in order to encode information**
 - Carrier wave: Basic sound wave transmitted through the circuit (provides a base which we can deviate)
- **Basic ways to modulate a carrier wave:**
 - Amplitude Modulation (AM)
 - Also known as Amplitude Shift Keying (ASK)
 - Frequency Modulation (FM)
 - Also known as Frequency Shift Keying (FSK)
 - Phase Modulation (PM)
 - Also known as Phase Shift Keying (PSK)

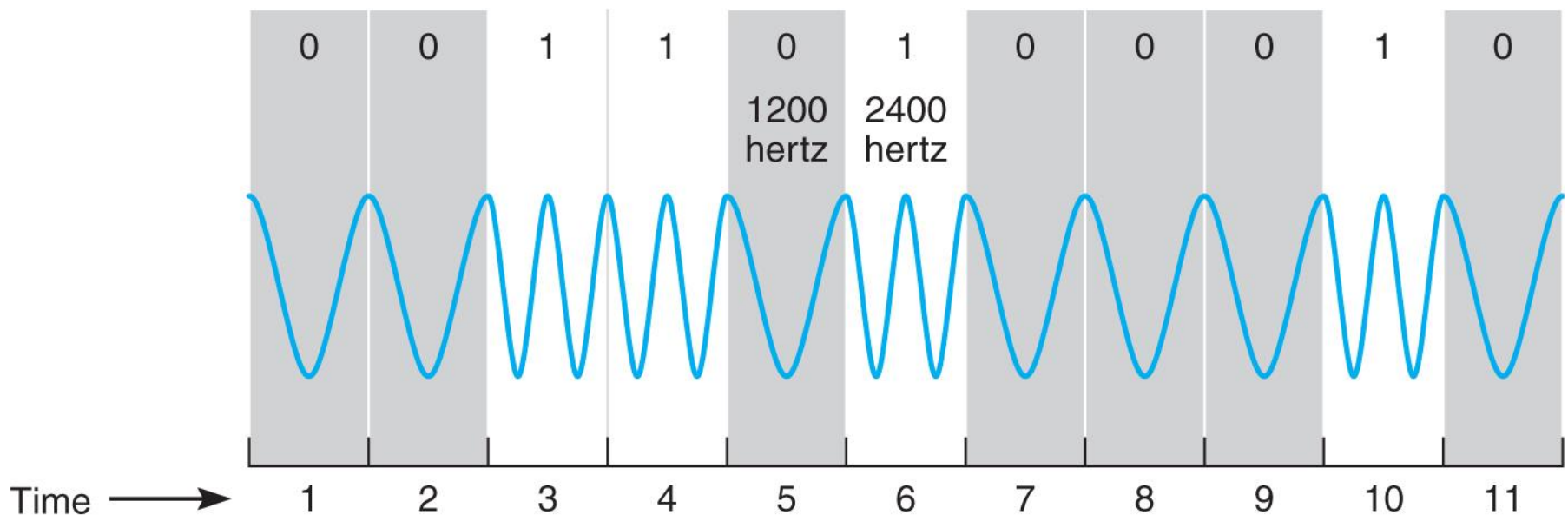
Amplitude Modulation (AM)

- Changing the height of the wave to encode data
- One bit is encoded for each carrier wave change
- More susceptible to noise



Frequency Modulation (FM)

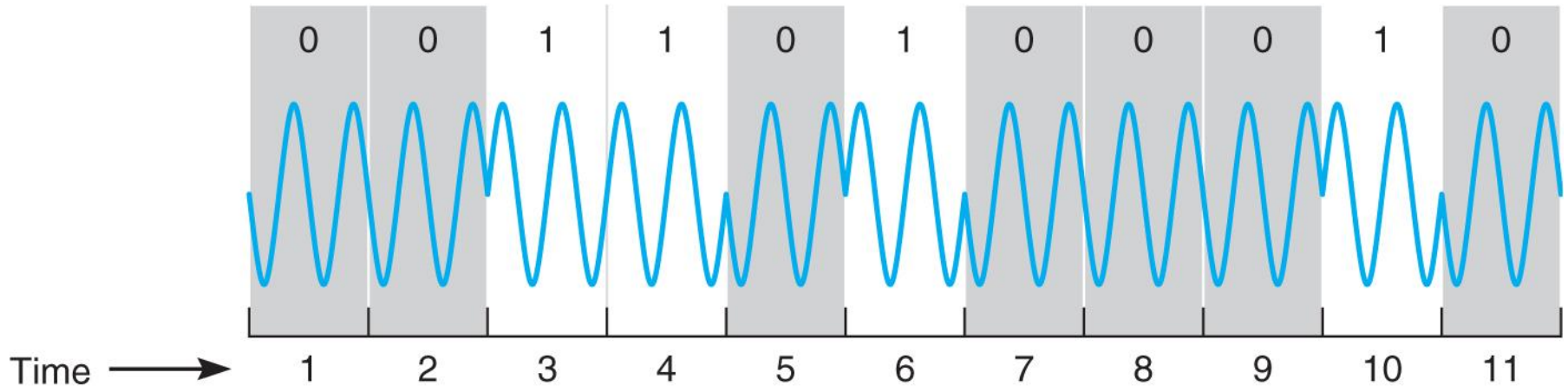
- Changing the frequency of carrier wave to encode data
- One bit is encoded for each carrier wave change



Phase Modulation (PM)

- Changing the phase of the carrier wave to encode data
- One bit is encoded for each carrier wave change
- Changing carrier wave's phase by 180° corresponds to a bit value of 1
- No change in carrier wave's phase means a bit value of 0

Phase Modulation (PM)



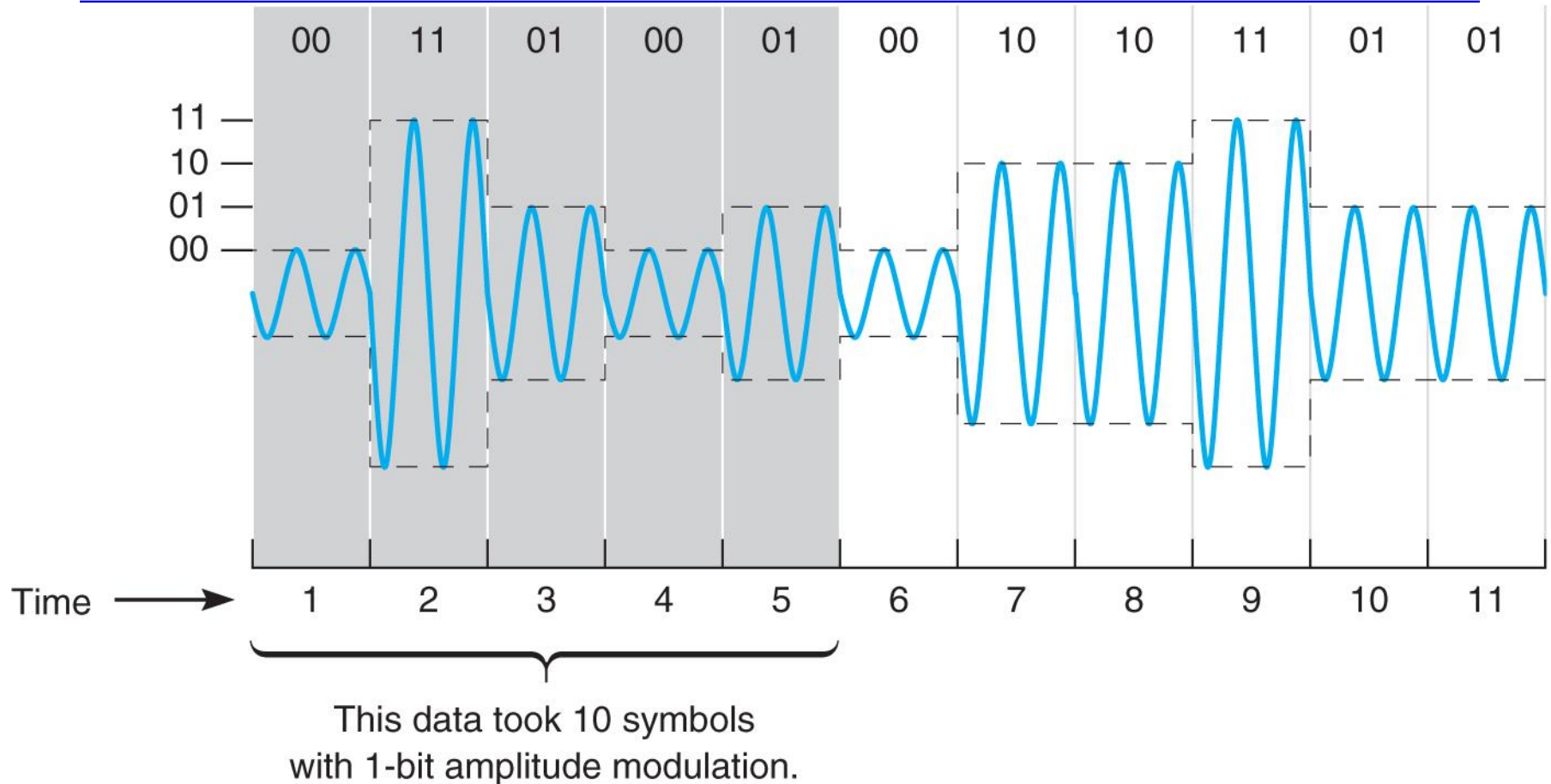
Concept of Symbol

- **Symbol: Use each modification of the carrier wave to encode information**
- **Sending one bit of information at a time**
 - One bit encoded for each symbol (carrier wave change) → 1 bit per symbol
- **Sending multiple bits simultaneously**
 - Multiple bits encoded for each symbol (carrier wave change) → n bits per symbol, $n > 1$
 - Need more complicated information coding schemes

Sending Multiple Bits per Symbol

- **Possible number of symbols must be increased**
 - 1 bit of information → 2 symbols
 - 2 bits of information → 4 symbols
 - 3 bits of information → 8 symbols
 - 4 bits of information → 16 symbols
 - n bits of information → 2^n symbols
- **Multiple bits per symbol might be encoded using amplitude, frequency, and phase modulation**
 - e.g., PM: phase shifts of 0° , 90° , 180° , and 270°
- **Subject to limitations: As the number of symbols increases, it becomes harder to detect**

Example: Two-bit AM



Bit Rate vs. Baud Rate or Symbol Rate

- Bit: a unit of information
- Baud: a unit of signaling speed
- Bit rate (or data rate): **b**
 - Number of bits transmitted per second
- Baud rate or symbol rate: **s**
 - number of symbols transmitted per second
- General formula:

$$b = s \times n$$

where

b = Data Rate (bits/second)

s = Symbol Rate (symbols/sec.)

n = Number of bits per symbol

Example: AM

$$n = 1$$

$$\rightarrow b = s$$

Example: 16-QAM

$$n = 4$$

$$\rightarrow b = 4 \times s$$

Bandwidth of a Voice Circuit

- **Difference between the highest and lowest frequencies in a band or set of frequencies**
- **Human hearing frequency range: 20 Hz to 14 kHz**
 - **Bandwidth = $14,000 - 20 = 13,800$ Hz**
- **Voice circuit frequency range: 0 Hz to 4 kHz**
 - **Designed for most commonly used range of human voice**
- **Phone lines transmission capacity is much bigger**
 - **1 MHz for lines up to 2 miles from a telephone exchange**
 - **300 kHz for lines 2-3 miles away**

Data Capacity of a Voice Circuit

- Fastest rate at which you can send your data over the circuit (in bits per second)
 - Calculated as the bit rate: $b = s \times n$
 - Depends on modulation (symbol rate)
 - Max. Symbol rate = bandwidth (if no noise)
- Maximum voice circuit capacity:
 - Using QAM with 4 bits per symbol ($n = 4$)
 - Max. voice channel carrier wave frequency: 4000 Hz = max. symbol rate (under perfect conditions)
 - Data rate = $4 * 4000 \rightarrow 16,000$ bps
 - A circuit with a 10 MHz bandwidth using 64-QAM could provide up to 60 Mbps.

Data Compression in Modems

- **Used to increase the throughput rate of data by encoding redundant data strings**
- **Example: Lempel-Ziv encoding**
 - **Used in V.44, the ISO standard for data compression**
 - **Creates (while transmitting) a dictionary of two-, three-, and four-character combinations in a message**
 - **Anytime one of these patterns is detected, its index in dictionary is sent (instead of actual data)**
 - **Average reduction: 6:1 (depends on the text)**
 - **Provides 6 times more data sent per second**

3.6 Digital Transmission of Analog Data

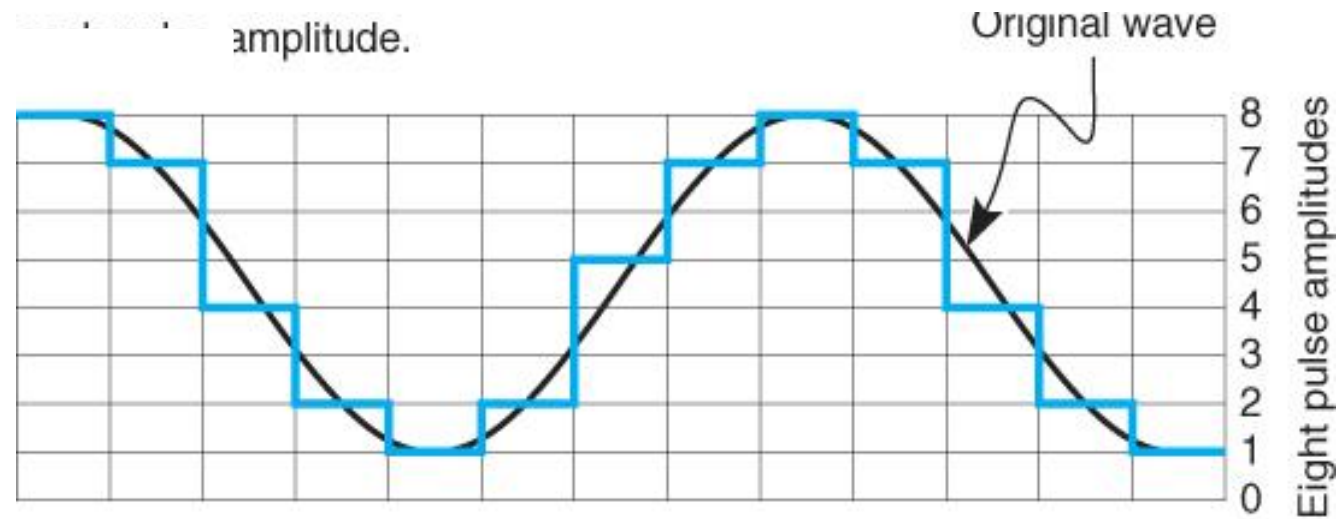
- Analog voice data sent over digital network using digital transmission
- Requires a pair of special devices called Codec - Coder/decoder
 - A device that converts an analog voice signal into digital form
 - Converts it back to analog data at the receiving end
 - Used by the phone system
- Modem is reverse device than Codec, and this word stands for Modulate/Demodulate. Modems are used for analog transmission of digital data.



Analog to Digital Conversion

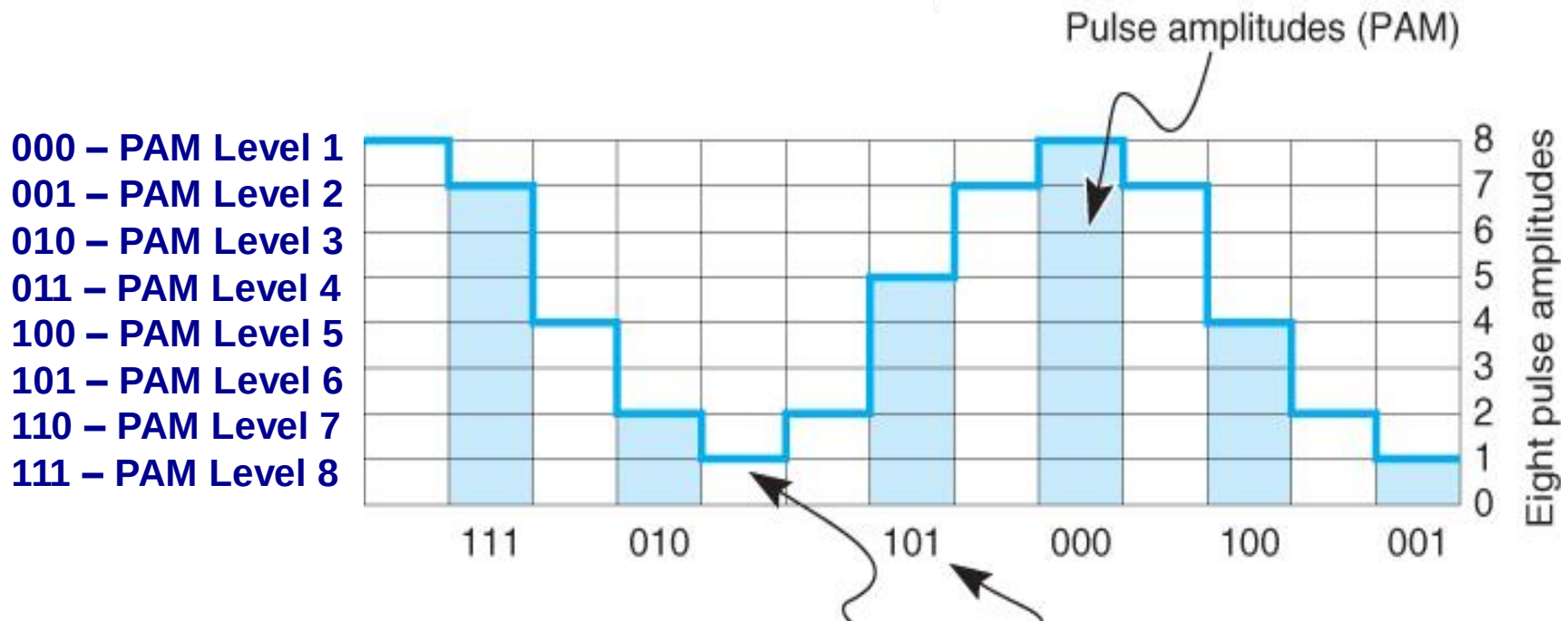
- Analog data must be translated into a series of bits before transmission onto a digital circuit
- Done by a technique called Pulse Amplitude Modulation (PAM) involving 4 steps:
 1. Take samples of the continuously varying analog signal across time
 2. Measure the amplitude of each signal sample
 3. Encode the amplitude measurement of the signal as binary data that is representative of the sample
 4. Send the discrete, digital data stream of 0's and 1's that approximates the original analog signal
- Creates a rough (digitized) approximation of original signal
 - Quantizing error: difference between the original analog signal and the replicated but approximated, digital signal
 - The more samples taken in time, the less quantizing error

PAM – Measuring Signal



- Sample analog waveform across time and measure amplitude of signal
- In this example, quantize the samples using only 8 pulse amplitudes or levels for simplicity
- Our 8 levels or amplitudes can be depicted digitally by using 0's and 1's in a 3-bit code, yielding 2^3 possible amplitudes

PAM – Encoding and Sampling

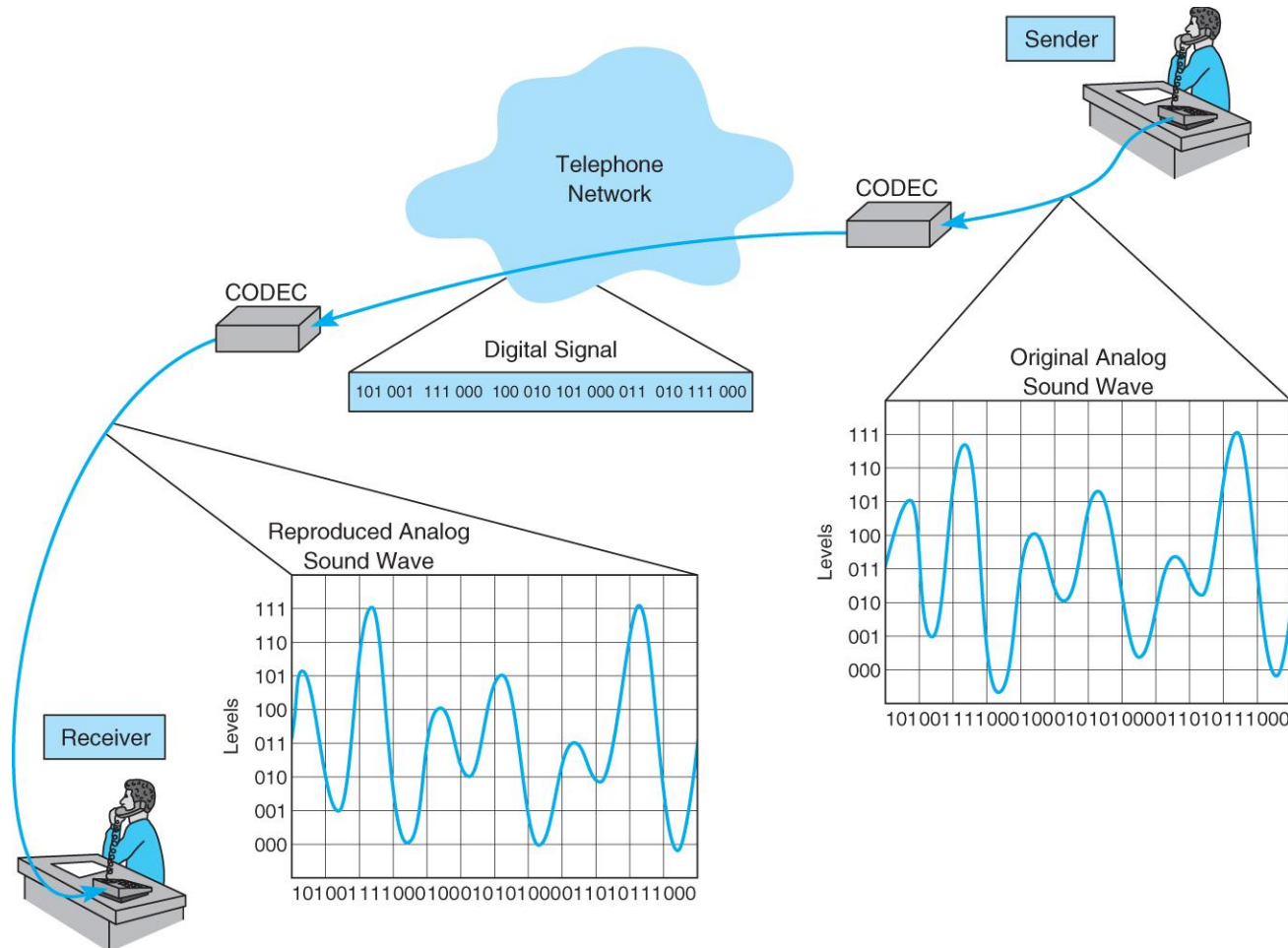


- For digitizing a voice signal, it is typically 8,000 samples per second and 8 bits per sample
- 8,000 samples x 8 bits per sample → 64,000 bps transmission rate needed
- 8,000 samples then transmitted as a serial stream of 0s and 1s

Minimize Quantizing Errors

- **Increase number of amplitude levels**
 - Difference between levels minimized → smoother signal
 - Requires more bits to represent levels → more data to transmit
 - Adequate human voice: 7 bits → 128 levels
 - Music: at least 16 bits → 65,536 levels
- **Sample more frequently**
 - Will reduce the length of each step → smoother signal
 - Adequate Voice signal: twice the highest possible frequency (4Khz x 2 = 8000 samples / second)
 - RealNetworks: 48,000 samples / second

PAM for Telephones



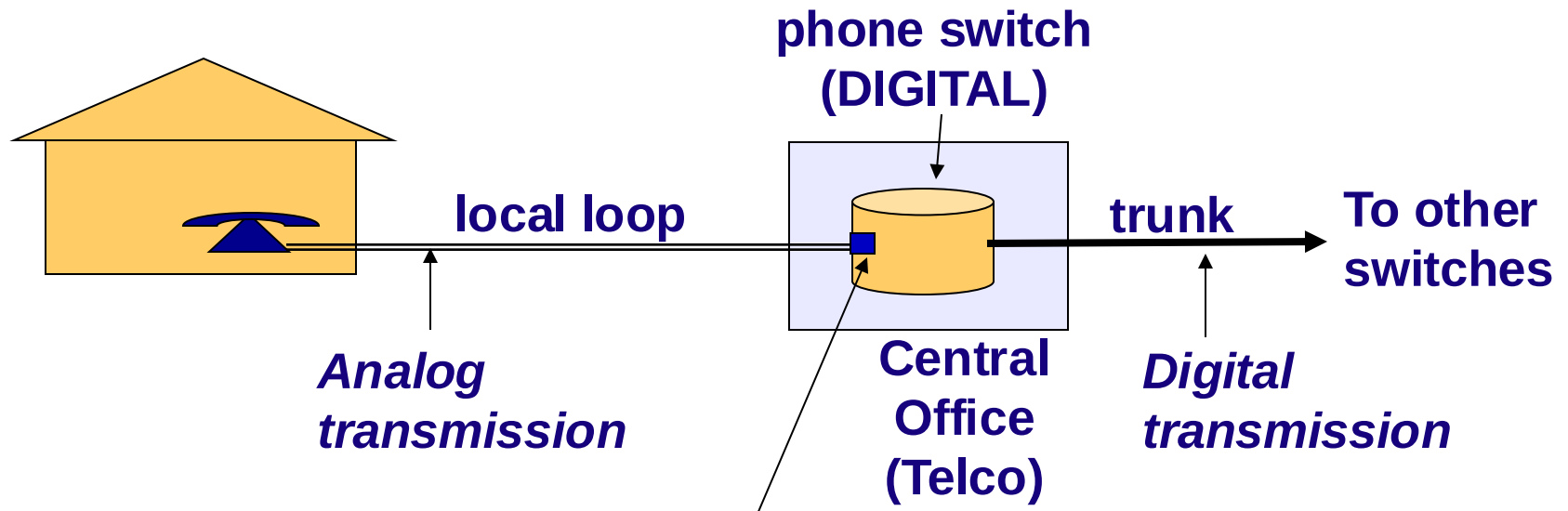
Combined Modulation Techniques

- **Combining AM, FM, and PM on the same circuit**
- **Examples**
 - **QAM - Quadrature Amplitude Modulation**
 - **A widely used family of encoding schemes**
 - **Combine Amplitude and Phase Modulation**
 - **A common form: 16-QAM**
 - **Uses 8 different phase shifts and 2 different amplitude levels**
 - » **16 possible symbols → 4 bits/symbol**
 - **TCM – Trellis-Coded Modulation**
 - **An enhancement of QAM**
 - **Can transmit different number of bits on each symbol (6,7,8 or 10 bits per symbol)**

Name	Symbol	Frequency	Wavelength	Applications
Extremely low frequency	ELF	3–30 Hz	10,000–100,000 km	Directly audible when converted to sound, communication with submarines
Super low frequency	SLF	30–300 Hz	1,000–10,000 km	Directly audible when converted to sound, AC power grids (50–60 Hz)
Ultra low frequency	ULF	300–3000 Hz	100–1,000 km	Directly audible when converted to sound, communication with mines
Very low frequency	VLF	3–30 kHz	10–100 km	Directly audible when converted to sound (below ca. 20 kHz; or <i>ultrasound</i> otherwise)
Low frequency	LF	30–300 kHz	1–10 km	AM broadcasting, navigational beacons
Medium frequency	MF	300–3000 kHz	f 100–1000 m	Navigational beacons, AM broadcasting, maritime and aviation communication

Name	Symbol	Frequency	Wavelength	Applications
High frequency	HF	3–30 MHz	10–100 m	Shortwave, amateur radio, citizens' band radio
Very high frequency	VHF	30–300 MHz	1–10 m	FM broadcasting, amateur radio, broadcast television, aviation, Group Penetrate Radar(GPR)
Ultra high frequency	UHF	300–3000 MHz	10–100 cm	Broadcast television, amateur radio, mobile telephones, cordless telephones, wireless networking, remote keyless entry for automobiles, microwave ovens, GPR
Super high frequency	SHF	3–30 GHz	1–10 cm	Wireless networking, satellite links, microwave links, satellite television, door openers
Extremely high frequency	EHF	30–300 GHz	1–10 mm	Microwave data links, radio astronomy, remote sensing, advanced weapons systems, advanced security

PCM - Pulse Code Modulation



- DS-0 is the basic digital communications unit used by phone network
- DS-0 corresponds to 1 digital voice signal

convert analog signals to digital data using PCM (similar to PAM)

- 8000 samples per second and 8 bits per sample (7 bits for sample + 1 bit for control)

→ 64 Kb/s (DS-0 rate)

ADPCM

- **Adaptive Differential Pulse Code Modulation**
- **Encodes the differences between samples**
 - The change between 8-bit value of the last time interval and the current one
 - Requires only 4 bits since the change is small
 - Only 4 bits/sample (instead of 8 bits/sample),
 - Requires $4 \times 8000 = 32$ Kbps (half of PCM)
 - Makes it possible to for IM to send voice signals as digital signals using modems (which has <56 Kbps)
- **Can also use lower sampling rates, at 8, 16 kbps**
 - Lower quality voice signals.

VoIP

- **Data and voice**
 - **Convergence into one network**
 - **Combine into one department**
- **Implementation**
 - **One set**
- **Costing**

3.7 Implications for Cyber Security

- **How secure the physical media**
 - Cables tap
 - Network equipment such as switches and routers in the data center
- **WiFi**
 - Signal is in the air.
- **Data Center**
 - Secure
 - Business continuity
 - DR and DC sites
 - Planning
 - Testing

References

- **Business Data Communications and Networking 14th Edition**
Jerry Fitzgerald, Alan Dennis, Alexandra Durcikova; Wiley
- www.Wikipedia.com

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Exercise Chapter 3

- E, F
- E Unipolar, Bipolar, Manchester encoding, Bipolar RTZ
- Paper 1 week